NEO Mathematics Learner Guide

Pythagoras and the Hidden Length - 10-session journey

The lesson should feel familiar even when the mathematics is new. Every session uses the same NEO lesson shell. Open one Cornerstone panel at a time, follow the 'What to do' steps, notice the mathematical goal, and carry the bridge into the next stage.

How to move through a lesson

- Read the Driving Question and Stepping Stones near the top.
- Open a Cornerstone panel. Begin with the numbered 'What to do' steps.
- Use the Scratchpad when drawing, calculating or organising ideas would help.
- Open the Practice Companion or Practice Studio when directed. Use Socratic prompt, Hint, Check and Worked solution in that order as needed.
- Pause. Return. Re-enter from the 'What to carry forward' bridge.
- Complete the Final Task in a format that communicates your mathematics clearly.

The 10-session journey

Session	Inquiry	Main movement
1	The Harmony of Numbers	Notice connections between number, music, shape and pattern.
2	What can three squares reveal?	Investigate the square-area relationship behind the theorem.
3	How can a theorem reveal a hidden length?	Use side roles to find a hypotenuse or shorter side.
4	Hidden Lengths Practice Studio	Build reliable method choice through 16 progressive questions.
5	Can we measure what we cannot reach?	Model real situations and ask whether Pythagoras is allowed.
6	How can one proof speak for every right triangle?	Explore Perigal's dissection and the nature of proof.
7	When harmony breaks	Meet square root of 2 and follow a proof by contradiction.
8	Are Pythagorean triples random?	Investigate families, scaling and a triple-generating machine.
9	Are squares really essential?	Test other similar shapes and identify the deeper area structure.
10	What makes Pythagoras more than a formula?	Build a final evidence exhibition across the whole unit.

Scratchpad reminder

The background is a mathematical reference surface. Erasing learner marks should not erase a square grid, coordinate grid, ratio table or number line. Use the Line tool when a straight side, axis, diagonal or construction line matters.

The six Cornerstones

Connection reveals relationships. **Movement** lets you change something and observe a mathematical response. **Creativity** asks you to make or communicate mathematics. **Reflection** uses evidence, errors and strategy. **Rest** supports pause and re-entry. **Nutrition** asks what the learning is nourishing practically or intellectually.

Mastery language

Emerging -> Developing -> Secure -> Mastering. These levels describe current evidence, not your worth or fixed ability. Reflection is part of mathematics: noticing what helps is useful data.

Final evidence

The unit ends with the question: **What makes Pythagoras more than a formula?** You will connect Harmony, Relationship, Proof, Disruption and Utility. Save useful Scratchpad pages and Final Tasks as you move through the unit.

A low-pressure route

You do not need to complete every optional extension. Start with the visible next step. Use a hint when it helps. A pause is allowed. A worked solution can be a re-entry tool. The mathematics will still be there when you return.